AN ACTIVE AUSTRALIAN SCIENCE 10 CURRICULUM MODULE

AUS BS10 SU STRAND: MODULE 2: THEORY OF BIOLOGICAL EVOLUTION

STUDENT'S NAME:		**TUTORIAL:**	
TEACHER'S NAME:		**DATE:**	

AUS Biological Sciences 10: Science Understanding Strand Module 2: Key Conceptual Understandings
• Students understand that within any single population there exist small differences between the characteristics of different individuals of that species; with such differences referred to as **variation**.
• Students understand that a **population** is the number of individual organisms of the same species living in a particular place at a given time.
• Students know that a **species** is a group of organisms that is genetically isolated from all other groups; i.e., two different species are incapable of exchanging genes by successfully interbreeding in nature.
• Students understand that variation results from the interaction between different genotypes of individual organisms with the environmental factors existing in their particular habitat.
• Students understand the difference between genotype and phenotype. **Genotype** refers to the actual alleles an individual possesses (or genetic make-up) that determines a specific trait. **Phenotype** refers to the characteristics of that specific trait as displayed by the individual. An organism's genotype and environment interact to produce that organism's phenotype.
• Students know that **natural selection** may be defined as differential reproductive success (fitness) of genotypes in a gene pool. In the context of evolution, "fitness" refers to the average reproductive output of a genotype in a gene pool.
• Students know that the sum total of alleles in a population is referred to as that population's **gene pool**.
• Students know that an **allele** is one of two or more forms of a gene situated in the same position (called a locus) on homologous chromosomes.
• Students know that once two gene pools become sufficiently different, then the resulting differences help to keep the two populations from exchanging genes through interbreeding, leading to **reproductive isolation**.
• Students understand that complete reproductive isolation together with natural selection and mutation can lead to increasing differences between populations, which over a long period of time, can result in the development of a new species.
• Student understand that populations change through the **process of natural selection** which may be summarised as follows: (1) Individuals in a population of a particular species show inherited variation, i.e., differences between individuals resulting from them having different alleles of a gene and different combinations of alleles of different genes. (2) In most species of organisms far more new individuals are produced each generation than survive to maturity, with only a small number of individuals surviving to produce offspring. (3) In a particular habitat subjected to environmental change, the inherited variation between individuals can result in some individuals of a species, by chance, having better characteristics to allow them to survive to maturity and reproduce. These better-adapted individuals consequently produce more offspring than the less well-adapted individuals. (4) Some of the offspring of these better-adapted parents will inherit the beneficial characteristics giving them a better chance to survive to maturity and reproduce, and to pass on the beneficial characteristics to some of their offspring. With each new generation there will be a higher proportion of individuals with the beneficial characteristics, eventually leading to a whole population better adapted to the changed environmental factors.

• Students understand that biological diversity or **biodiversity** refers to the variety of all life forms, including the different plants, animals and microorganisms, the genes they contain and the ecosystems of which they are a part.
• Students know that an **ecosystem** is in the main a self-sustaining system formed by living organisms interacting with one another, and with their physical environment.
• Students understand that **artificial selection** is the process in which humans direct the differential reproductive success of organisms in a breeding population to yield a desired phenotype in future generations.
• Students weigh scientific evidence for biological evolution that comes from (1) Study of the genomes of species to see how much genetic similarity or difference there is between species and then comparing that with how closely related scientists think that the species are. (2) **Biogeographical studies:** Modern distribution patterns of biota are basically similar to those of fossilised relatives, because modern animals and plants have descended from ancestors living in the same areas. (3) **Comparing embryonic forms**: Organisms in the course of their embryonic development often develop structures that resemble those seen in ancestral adults. (4) **Studying the development of embryos:** Studies of how tissues develop in embryos shows that organisms retain a much greater capacity for developing different structures than their actual form suggests. (5) **Anatomical studies:** Many organisms share basic homologous structural features, even though in external form they may look different. (6) **Study of intermediate forms:** Both fossils and modern organisms provide ample evidence for the transition of one kind of organism into another.

An Introduction to Evolution

In 1973, Theodosius Dobzhansky published his most famous essay titled: “Nothing in biology makes sense except in the light of evolution.”

1. What is biological evolution? How would you explain its meaning to your friends?

[**2 marks**]

2. What six primary sources of scientific data provide evidence that biological evolution has occurred?

[**6 marks**]

3. In 1844, a controversial conclusion of the theory of biological evolution, particularly human evolution, was that we humans share a common ancestor with the great apes. Do you agree that we share a common ancestor with apes? If you don't agree, why don't you agree?

[**2 marks**]

4. Why should the Theory of Biological Evolution be taught in all Australian schools?

[**2 marks**]

5. Why should creationism and intelligent design not be taught in Australian science classrooms?

[**2 marks**]

The Theory of Biological Evolution

The theory of biological evolution is a working hypothesis, based upon scientific evidence, which tries to explain observed changes to living things over time. Biological evolution is a process that results in heritable changes in a population spread over many generations.

- Scientific Concept: Biological evolution can be more precisely defined as any change in the frequency of alleles within a gene pool from one generation to the next.

Human beings and many other species can have no more than two alleles for a given gene, each allele residing on a separate, homologous chromosome. Many allelic variants of a gene can, however, exist in a population. The sum total of alleles in a population is referred to as that population's **gene pool**.

- Science Concept: A species is a group of organisms that is genetically isolated from all other groups; i.e., two different species are incapable of exchanging genes by successfully interbreeding in nature.

- Science Concept: An allele is one of two or more forms of a gene situated in the same position (called a locus) on homologous chromosomes.

- Scientific Concept: A population is the number of individual organisms of the same species living in a particular place at a given time.

Key Principles of Biological Evolution

A first key principle of biological evolution is that of descent with modification. This principle describes the process by which species of living things can undergo modification over time, with such change sometimes resulting in the formation of new, separate species. All species on Earth have descended from other species, and a single, common ancestor lies at the base of the evolutionary tree.

A second key principle of biological evolution is that of natural selection. Natural selection is a process in which the differential adaptation of organisms to their environment selects those traits that will be passed on, (by reproduction), with greater frequency from one generation to the next. Importantly, natural selection acts on individuals, but it is populations that evolve.

Variation and Natural Selection

- Science Concept: Within any single population there exist small differences between the characteristics of different individuals of that species; with such differences referred to as **variation**.

- Science Concept: Variation results from the interaction between the different genotypes of individual organisms with the environmental factors existing in their particular habitat.

- Science Concept: A population is the number of individual organisms of the same species living in a particular place at a given time.

- Science Concept: A species is a group of organisms that is genetically isolated from all other groups; i.e., two different species are incapable of exchanging genes by successfully interbreeding in nature.

The Peppered Moth

Peppered moths can occur naturally as two differently coloured forms, a typical light-coloured form known as typicals or (*typica*), and a dark-coloured form known as melanics or (*carbonaria*) as shown in the images on the next page.

- Science Concept: Melanism is the occurrence of a black phenotype in a normally light-coloured species brought about by genetic mutation.

In England, before the Industrial Revolution, the populations of peppered moth were mainly light-coloured moths, with some dark-coloured moths present. From the 1850's to the 1950's, increasing numbers of dark-coloured peppered moths were observed, while numbers of light-coloured moths were observed to decrease, especially in industrial areas, polluted with black soot. As anti-pollution legislation took effect, resulting in a cleaner environment, less numbers of dark-coloured moths were observed, while numbers of light-coloured moths increased.

- Science Concept: Humans and many other species can have no more than two alleles for a given gene, each allele residing on a separate, homologous chromosome.

- Science Concept: An allele is one of two or more forms of a gene situated in the same position (called a locus) on homologous chromosomes.

A single locus with two alternate alleles, on separate homologous chromosomes, essentially controls the colour of peppered moths (*Biston betularia*). The genotype of dark-coloured peppered moths can be **BB** or **Bb**. The genotype of light-coloured peppered moth is **bb**. Uppercase **B** represents the allele for dark-colour, and lowercase **b** represents the allele for light-colour, in the heterozygote. The dominant allele **B** masks the effect of the recessive allele **b** in the heterozygote. The effect of the recessive allele **b** is hidden by the effect of the dominant allele **B** in the heterozygote.

Light-coloured Peppered moth

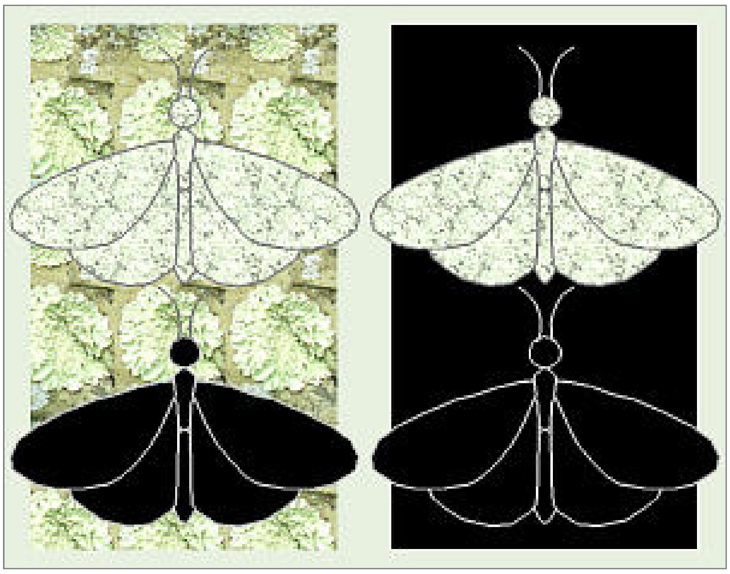
Dark-coloured Peppered moth

6. Can light-coloured peppered moths change into dark-coloured peppered moths and vice versa?

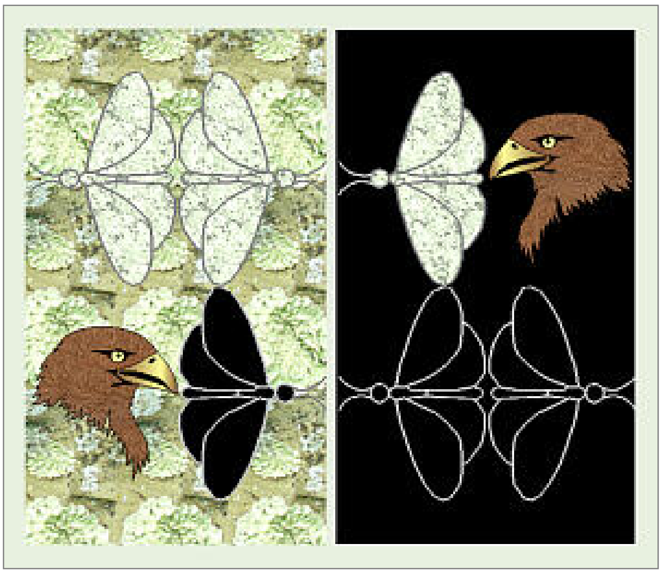
[**1 mark**]

7. Since peppered moths cannot change their colour, how then can you explain the observed change in the frequency of the light and dark-coloured forms of the peppered moth, as their environment changed from clean to soot polluted, and then soot polluted to clean?

[**4 marks**]



Peppered moths and lichens on trees



Peppered moths and predatory birds

8. Can you now explain the complete story of the change in phenotype of the peppered moth?

[**5 marks**]

Experimental Design of Peppered Moth Investigations

Has follow-up research revealed flaws in the experimental design used by H. B. D Kettlewell?

You will recall that the populations of peppered moths, in the cities of industrial England, went from mainly light-coloured moths to mainly dark-coloured moths, and then back again, as the pollution in their industrial environment increased and then decreased.

9. Is the change in colour (phenotype) of the peppered moth an example of biological evolution in action?

[**1 mark**]

The Process of Natural Selection

- Science Concept: Populations change through the **process of natural selection** which may be summarised as follows: (1) Individuals in a population of a particular species show inherited variation, i.e., differences between individuals resulting from them having different alleles of a gene and different combinations of alleles of different genes. (2) In most species of organisms far more new individuals are produced each generation than survive to maturity, with only a small number of individuals surviving to produce offspring. (3) In a particular habitat subjected to environmental change, the inherited variation between individuals can result in some individuals of a species, by chance, having better characteristics to allow them to survive to maturity and reproduce. These better-adapted individuals consequently produce more offspring than the less well-adapted individuals. (4) Some of the offspring of these better-adapted parents will inherit the beneficial characteristics giving them a better chance to survive to maturity and reproduce, and to pass on the beneficial characteristics to some of their offspring. With each new generation there will be a higher proportion of individuals with the beneficial characteristics, eventually leading to a whole population better adapted to the changed environmental factors.

- Science Concept: Natural selection may be defined as differential reproductive success (fitness) of genotypes in a gene pool. In the context of evolution, “fitness” refers to the average reproductive output of a genotype in a gene pool.

- Science Concept: The sum total of alleles in a population is referred to as that population’s gene pool.

- Science Concept: A population is the number of individual organisms of the same species living in a particular place at a given time.

10. Can you now explain the **process of natural selection** by referring to the change in phenotype of the peppered moth, from light-coloured to dark-coloured, and then back again, as soot pollution in their industrial environment increased and then decreased?

[**4 marks**]

The fast-evolving Deer Mouse

Deer mice are found in sandy soils in Nebraska. Over several thousand years, the deer mouse evolved a pale coat that helped it to evade predators like owls and hawks. Deer mice are widespread across North America, but they usually have dark coat, so that they are camouflaged in dark soils, and can stay hidden from raptors. Scientists at Harvard and at the University Of California, at Berkeley, discovered a single gene called the *Agouti* gene in light coloured deer mice, which is expressed in higher amounts and for longer than the genes that code for dark hair. The gene emerged about 4,000 years ago, which was only a few thousand years after the dark-coated mice first colonised their new sandy home. Its spread was rapid. Most animals that evolve new traits do so from variations of existing genes (alleles), but in deer mice, the light-coloured trait evolved from a completely new *Agouti* gene. The *Agouti* gene did not occur before colonisation of the sandy-soiled environment, and when it did appear, natural selection acted on it to confer the advantage of camouflage to the deer mice living in light-coloured soils.

- Science Concept: Mutations are sudden permanent changes in genes producing new alleles and genes.
- Science Concept: Mutants are individuals that show the effect of a mutation in their phenotype.

11. Do you think that the first light-coloured deer mouse (*Peromyscos maniculatus*) to have the *Agouti* gene was a **mutant**? You must fully explain your answer.

[**2 marks**]

Source of Image URL: http://news.bbc.co.uk/earth/hi/earth_news/newsid_8225000/8225219.stm

Antibiotic Resistance

- Antibiotic resistance is a drug resistance where a microorganism like a bacterium is able to survive exposure to an antibiotic.

Antibiotic resistance is an excellent example of natural selection 'in action'. Bacteria that acquire a mutation that allows them to survive an antibiotic will live long enough to reproduce and so spread the 'survival' gene to subsequent generations of that bacterium.

The basis of evolution is a change in the frequency of alleles within a population (a phenomenon that includes the appearance of new alleles through mutation). Evolution at this level is called microevolution.

Mechanisms of Microevolution

- Science Concept: Microevolution can be precisely defined as any change in the frequency of alleles in a gene pool from one generation to the next generation.
- Science Concept: The sum total of alleles in a population is referred to as that population's gene pool.
- Science Concept: A population is the number of individual organisms of the same species living in a particular place at a given time.

The five (5) evolutionary mechanisms that can change the frequency of alleles in a gene pool are: mutation, gene flow, genetic drift, nonrandom mating, and natural selection.

(1) **Mutation**

Mutation happens fairly infrequently, and most mutations either have no effect or are harmful. However adaptive mutations are vital for evolution in that they are the only means by which entirely new alleles and new genes are produced.

(2) **Gene Flow**

Gene flow**,** also called gene migration, is the introduction of alleles (by interbreeding) from one population of a species to another, thereby changing the composition of the gene pool of the receiving population. The introduction of new alleles through gene flow increases variability within the population and makes possible new combinations of traits. In human beings gene flow usually comes about through the actual migration of human populations, either voluntary or forced.

(3) **Genetic Drift**

Genetic drift, the chance alteration of allele frequencies in a population, has its greatest effects on small populations. Genetic drift operates on small populations in two ways. The first of these is the bottleneck effect, defined as a change in allele frequencies due to chance during a sharp reduction in a population's size. The second is the founder effect: the fact that when a small subpopulation migrates to a new area to start a new population, it is likely to bring with it only a portion of the original population's gene pool.

(4) **Non-Random Mating**

Non-random mating is mating in which a given member of a population is not equally likely to mate with any other member. Sexual selection is a form of non-random mating that can directly affect the frequency of alleles in a gene pool. It occurs when differences in reproductive success arise because of differential success in mating.

(5) **Natural Selection**

Some individuals will be more successful than others in surviving and hence reproducing, owing to traits that better adapt them to their environment. This phenomenon is known as natural selection. Natural selection is the only mechanism of microevolution that consistently operates to adapt organisms to their environments. Therefore, natural selection is regarded as the most powerful mechanism producing biological evolution.

Macroevolution and Speciation

- Science Concept: Macroevolution may be considered as cumulative microevolution that results in the formation of a new species from a pre-existing species, with each organism linked via descent to a common ancestor.

- Science Concept: Speciation is the process of a single species becoming two or more species. Biologists know little about the genetic mechanisms of speciation. How species originate is still largely a biological mystery.

- Science Concept: Once two gene pools become sufficiently different; then the resulting differences help to keep the two populations from exchanging genes through interbreeding, which results in reproductive isolation.

- Science Concept: Complete reproductive isolation together with mutation and natural selection can produce increasing differences between two populations, which over a long period of time, can result in the development of a new species.

Speciation and Biological Diversity

Speciation ⟶ Increasing Biological Diversity

Extinction is the ultimate fate of all species.

Extinction ⟶ Decreasing Biological Diversity

- Science Concept: Biological diversity or biodiversity refers to the variety of all life forms, including the different plants, animals and microorganisms, the genes they contain, and the ecosystems of which they are a part.

- Science Concept: An ecosystem is in the main a self-sustaining system formed by living organisms interacting with one another, and with their physical environment.

- Science Concept: The diversity of life on Earth is due to the innate ability of plant and animal species to evolve, adapt, and develop within a variety of environmental (or ecological) niches and habitats.

- Science Concept: Mass extinction is followed by periods of radiation where new species evolve to fill the empty environmental or ecological niches left vacant by their previous occupiers.

- Science Concept: Ecological niche as a concept is multidimensional as it includes where a species lives and what it does, how it does it, and when it does it, etc. An ecological niche is the particular role played by the organisms of a species within an ecosystem.

Artificial Selection

- Science Concept: Artificial selection is the process in which humans direct the differential reproductive success of organisms in a breeding population to yield a desired phenotype in future generations.
- **Class Debate:** With your class teacher's approval, would you like to debate this Question: Is it morally wrong to artificially select humans?
    - Click YES: ☐ or Click NO: ☐

Scientific Evidence for Biological Evolution

- Science Concept: Scientific evidence for biological evolution comes from (1) Study of the genomes of species to see how much genetic similarity or difference there is between species and then comparing that with how closely related scientists think that the species are. (2) Biogeographical studies: Modern distribution patterns of biota are basically similar to those of fossilised relatives, because modern animals and plants have descended from ancestors living in the same areas. (3) Comparing embryonic forms: Organisms in the course of their embryonic development often develop structures that resemble those seen in ancestral adults. (4) Studying the development of embryos: Studies of how tissues develop in embryos shows that organisms retain a much greater capacity for developing different structures than their actual form suggests. (5) Anatomical studies: Many organisms share basic homologous structural features, even though in external form they may look different. (6) Study of intermediate forms: Both fossils and modern organisms provide ample evidence for the transition of one kind of organism into another.

Archaeopteryx: A transitional Fossil

Most palaeontologists believe that Archaeopteryx is a transitional fossil between dinosaurs and birds. Archaeopteryx fossils are therefore very important in the study of the origin of birds and of dinosaurs.

12. Why are Archaeopteryx fossils very important in the study of the origin of birds and dinosaurs?

[**4 marks**]

Source of image URL: http://en.wikipedia.org/wiki/Transitional_fossil

Extra Online Resources

Internet Use You may like to access these extra URL's to learn more about Biological evolution?

URL 1: http://phet.colorado.edu/en/simulation/natural-selection
URL 2: http://glencoe.mcgraw-hill.com/sites/dl/free/0078695104/383939/BL_12.html
URL 3: http://wps.prenhall.com/wps/media/objects/487/499482/CDA21_1/CDA21_1b/CDA21_1b.htm
URL 4: http://wps.prenhall.com/wps/media/objects/487/499482/CDA21_1/CDA21_1d/CDA21_1d.htm
URL 5: http://www.nature.com/scitable/knowledge/library/natural-selection-genetic-drift-and-gene-flow-15186648

Key Concepts

What new science ideas have you now learned about **Biological evolution** by studying this AUS BS10 SU Strand Module 2? List your newly learned key Science Concepts, as dot-points, in the textbox given below.

A Self-Check Summative Assessment Table		
Now complete this assessment table by adding the 'self-check' mark achieved for your answer to each focus question. Your teacher will provide his/her 'model' answer to each question, via 'whole of class' discussion.		
Total Mark Possible = **35**	Mark Achieved =	Percent (%) Score Achieved =